Arkesh Jayakumar

arkeshjayakumar@gmail.com | Hyderabad, Telangana, India
Portfolio: rkz-10.github.io | Blog: rkz-10.github.io/blog | GitHub: github.com/rkZ-10
LinkedIn: linkedin.com/in/arkesh-jayakumar-836866190

PROFESSIONAL SUMMARY

Lead Backend Engineer with 6+ years of experience at SysCloud building and scaling distributed, cloud-based backend systems on AWS, including sole ownership of an internal credential-authentication system serving 88 client services and a 185K-line security-compliance product across two SaaS platforms. Currently the product-logic authority (reviewing behavior and writing fix specs, not the implementer) on a modernization of the company's flagship security product led by a Europe-based Principal Architect, while coordinating regularly with engineering teams in the US and Japan. Works daily with AI coding assistants (Claude, GitHub Copilot).

CORE COMPETENCIES

Distributed Systems & Microservices • Cloud Architecture & AWS • Config-Driven Workflow Orchestration • Secrets & Credential Management • JWT & Diffie-Hellman Authentication • Multi-Tenant Architecture & Isolation • Compliance & PII Detection Systems • AI-Assisted Tooling & Automation • System Design & Technical Leadership • Cross-Functional Mentorship • Cross-Regional Engineering Collaboration

SKILLS

Languages: TypeScript, Go, PHP, Python, Java, C/C++

Backend: Node.js, Microservices Architecture, API Design & Development, Distributed Systems, JSON Config-Driven Task Orchestration

Cloud: AWS (S3, Lambda, DynamoDB, KMS, EC2, ECS, ECR, Bedrock, CloudWatch)

Databases: PostgreSQL, Microsoft SQL Server, Redis

Infrastructure: Docker, Lustre

Testing: Jest, PHPUnit

Observability: Grafana, Kibana

Security: JWT & Diffie-Hellman Authentication, Compliance/DLP Detection Systems, Phishing & Ransomware Detection

WORK EXPERIENCE

SysCloud Technologies Pvt Ltd

Lead Software Engineer Jun 2023 – Present

- Serving as product-logic authority for the **modernization of the legacy Security Product** from PHP to Go: validating each rebuilt component against original PHP behavior, identifying architectural and behavioral gaps, and producing implementation specs for the Principal Architect; retaining direct ownership of the Sync-triggered watch channel renewal, kept intentionally in PHP.
- Led design and rollout of **multi-tenant isolation in Cloud Mover**, closing a cross-business data-access gap across downloads, scans, and uploads: replaced a single shared IAM role with implicit access to every customer's S3 bucket with per-business roles scoped to just that tenant's bucket, and restructured the Lustre path hierarchy to segment storage by business ID directly under the Lustre ID.
- Upgraded the internal **Secrets Manager (SMv3)** from JWT-based to Diffie-Hellman authentication as part of a company-wide security hardening initiative, building a purpose-built client package for auth and data retrieval and extending multi-tenant support for KMS-encrypted, DynamoDB-stored role data; later extended it with cross-team service endpoints (MTIT role retrieval and GCP/Azure BYOK credential access) for teams building on top of SM. It now serves 195 registered clients (124 Lambdas, 71 ECS stacks) across 10 teams and 7 AWS regions, processing **~9M API requests/day at a 99.9%+ success rate**.

- Spearheaded **Backup Data Insights (BDI)**, porting Compliance and Ransomware Detection logic from PHP to TypeScript onto CM Runner Plus, a config-driven workflow engine extending the CM Runner framework (co-built with teammates): migrated 6 compliance policies onto it initially, then designed a natural-language-to-config generator that added 7 more with minimal manual effort.
- Designed and developed the **Excel-based Policy CRU system** across three Lambda functions. A shared capability table drives what each of the 9 policy types requires across generation, validation, and encoding, replacing roughly 50 duplicated type-check branches that had drifted independently across the codebase.
- Independently created **three AI-tooling contributions** outside assigned scope: a personal Azure DevOps MCP integration automating work reporting; a system prompt converting natural-language policy descriptions into validated CM Runner Plus configs (the generator behind the 7 additional compliance policies above); and a persistent Knowledge Base for the team's Claude-based debugging agent.
- **Directed and mentored a team of 7 engineers** across parallel workstreams: refactored the Bucket Encryption API, guided development of a standardized Lustre mount script now used company-wide, and oversaw a BDI Actions job, an eDiscovery indexing service, and a Lambda-based Drive/Mail preview service.

Senior Software Engineer Jul 2021 – May 2023

- Owned the **internal Secrets Manager API**, replacing its unauthenticated, self-asserted trust model with credential-validated, 1-hour-expiry JWTs re-verified on every request across 15 protected endpoints (brokering database connection strings, per-user encryption keys, and service-account credentials to 88 registered client services across the backup, data-access, and unified-backup platforms); later added zero-downtime credential rotation, used to cycle 76 of 79 client credentials without service interruption. It now handles ~4.5M API requests/day at a 99.999%+ success rate (24 failures).
- **Sole owner of the legacy Security Product** (185K-line PHP codebase) across Google Workspace and Microsoft 365, encompassing near-real-time compliance/PII content scanning (16 classifiers spanning HIPAA, PCI, FERPA, CIPA and more, plus 19 built-in PII pattern types), sharing/access-control violation detection (17 violation types), login/Drive-activity anomaly detection (~2-second worker latency), Phishing and Ransomware detection, and App Firewall enforcement.
- **Founding engineer on Cloud Mover**, a microservice extracting core backup data-movement logic out of the monolithic Backup product; helped build and iteratively evolve CM Runner (a declarative JSON-workflow orchestration framework designed by the CTO) alongside teammates during development, with per-item partial failures isolated via reflect-and-continue semantics, and observability handled through Grafana dashboards drawing on Kibana log indexes, a Grafana timeseries DB, and AWS CloudWatch metrics, alongside per-service container logs in Kibana.

Software Engineer Jul 2020 – Jun 2021

- Built **Phishing and Ransomware detection capabilities** for the legacy Security Product, which at the time already provided real-time compliance/student-safety content scanning, App Firewall enforcement, and Drive/Login-activity anomaly detection.
- Ramped up on the internal Secrets Manager service through knowledge transfer and incident-response support, ahead of taking full ownership of its API as Senior Software Engineer.

Intern Jan 2020 – Jun 2020

- Onboarded onto the Security Product's architecture and day-to-day operations ahead of transitioning into ownership as Software Engineer.

PROJECTS

Travel Assistant github.com/rkZ-10/travel-assistant

- Built an AI travel assistant around a **9-tool MCP server**, with **hybrid RAG over official airline and DGCA pages** that returns citations, and a **Claude Agent SDK planner** whose guardrails are enforced in code.
- Added a React + FastAPI UI and a **23-case evaluation suite** that passes 23/23.

Laravel Blog rkz-10.github.io/blog

- Built a blog on **Laravel 13** with MariaDB, Redis, queued jobs and Pest tests, exported as a static site to GitHub Pages through GitHub Actions.

EDUCATION

B.Tech, Computer Science Engineering, SRM University 2020
Chennai, India